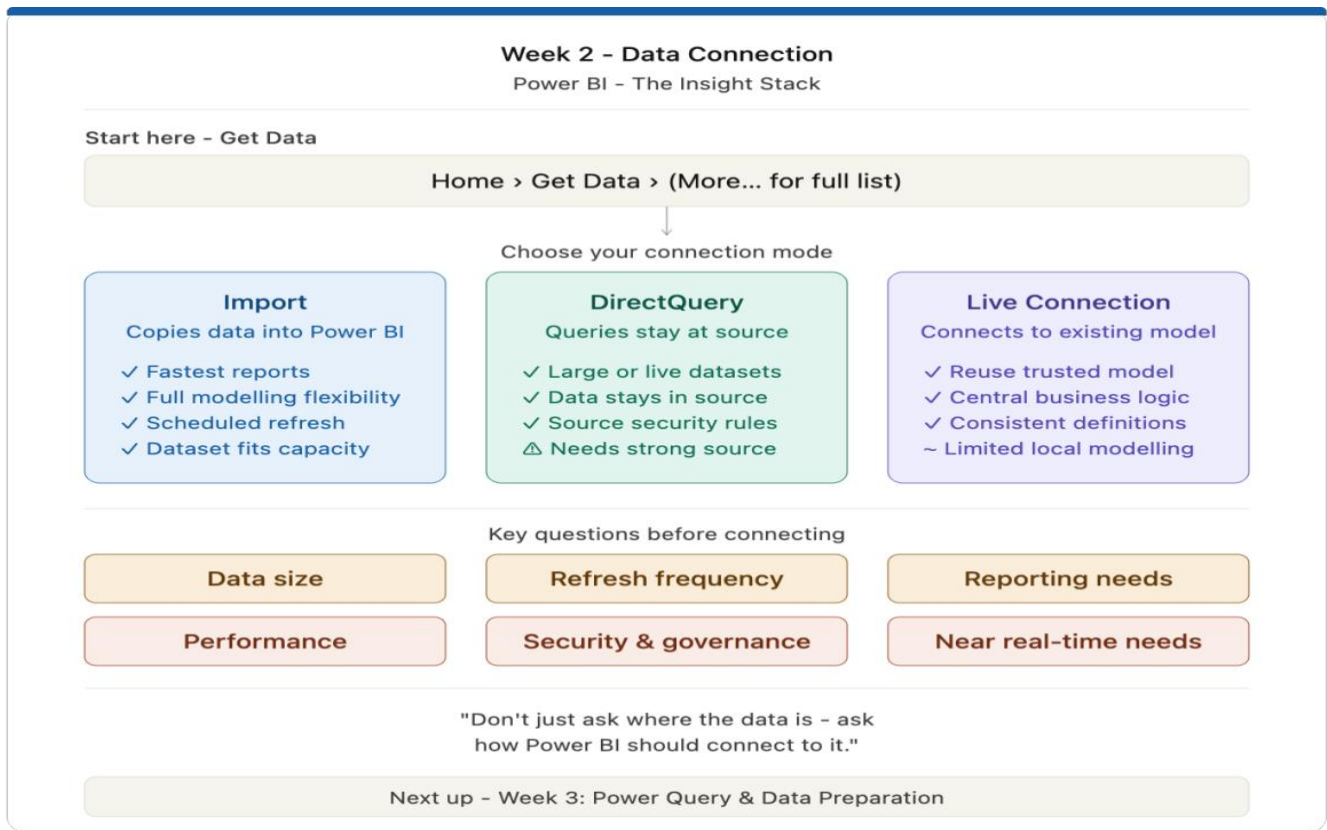




Why Data Connection Matters

Before a Power BI report can answer business questions, it needs access to the right data. Data connection is the process of linking Power BI to the sources where business data is stored. These sources may include Excel files, databases, SharePoint folders, cloud platforms, Microsoft Fabric, online services, or existing semantic models.

In Power BI Desktop, this usually begins with: **Home > Get Data**

If the required source is not visible under common data sources, selecting **More** opens the full Get Data window.

Data Source Categories in Power BI

Data Source category	
ALL	The All category includes all data connection types from all categories.
File	Connects to local or cloud-based files like Excel, CSV, JSON, and PDF.
Database	Connects to structured databases such as SQL Server, MySQL, Oracle, and PostgreSQL.
Microsoft Fabric	Connects to Microsoft Fabric data services including Lakehouse, Warehouse, and <u>OneLake</u> .
Power Platform	Uses Power Platform sources like Dataverse and Power BI dataflows.
Azure	Connects to Azure cloud services such as Azure SQL, Blob Storage, and Synapse.
Online Services	Connects to online apps and SaaS platforms like SharePoint, Salesforce, and Google Analytics.
Other	Includes miscellaneous and advanced connectors not grouped elsewhere.

A useful note for analysts: some connectors may be in **Preview** or **Beta**, which means they may have limited support and should be used carefully in production environments.

Further reading: [Microsoft Learn – Power Query connectors](#)

The Bigger Question: How Should Power BI Connect?

In Power BI, the three main connection approaches are:

- **Import**
- **DirectQuery**
- **Live Connection**

There are also more advanced approaches such as **Direct Lake**, **Hybrid Tables**, and **Composite Models**, but the three core connection types are the best starting point.

Import Mode :

It is the default and most common connection method in Power BI.

In this method, Power BI loads a copy of the selected data into the semantic model. Reports then query Power BI's in-memory engine instead of repeatedly querying the original data source. The main **benefit** of Import mode is **speed and flexibility**.

For most standard business reporting scenarios, **Import should be the starting point** unless there is a clear reason to use another connection method.

DirectQuery Mode

DirectQuery keeps the data in the original source.

Instead of importing a full copy into Power BI, Power BI sends queries back to the source whenever users interact with the report. However, DirectQuery needs careful design.

It can create slower reports if the source system, network, query design, or gateway setup is not strong enough. Some features may also be limited compared to Import mode.

Live Connection

A **Live Connection** connects Power BI to an existing semantic model, such as a published Power BI semantic model or Analysis Services model. The key difference is this:

DirectQuery connects to a data source. Live Connection connects to an already-built model.

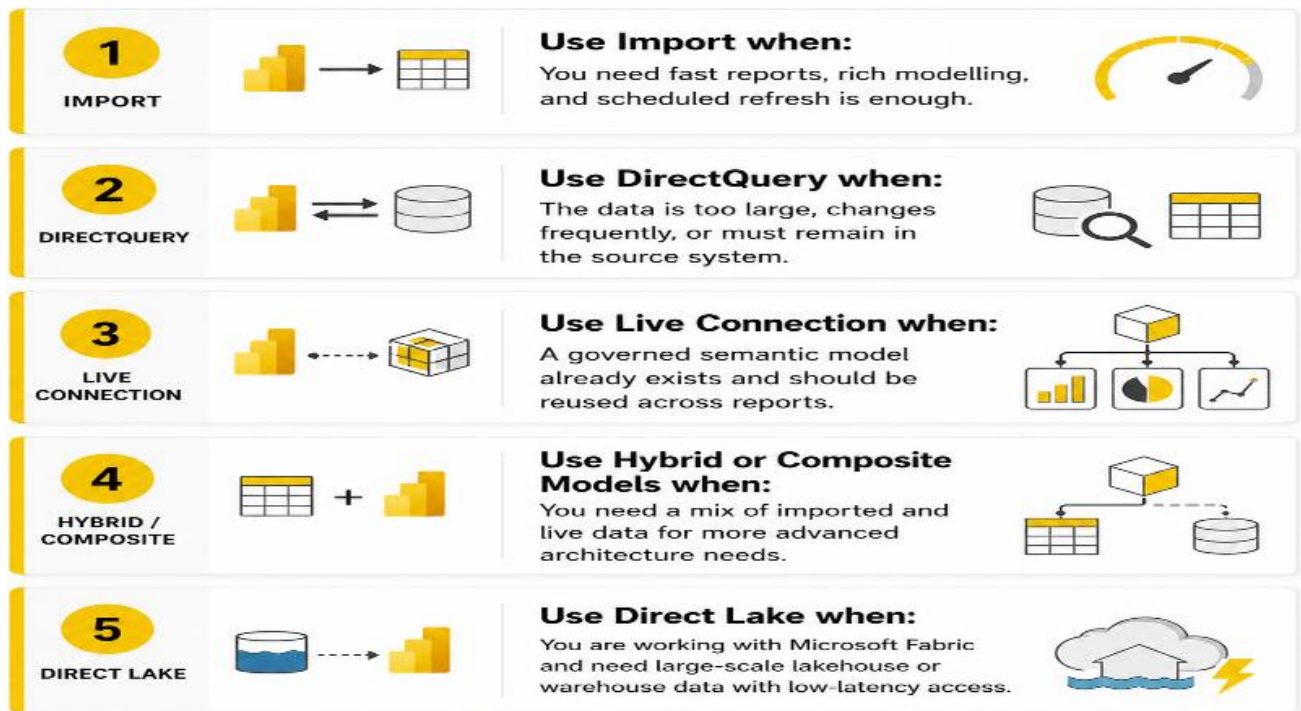
The trade-off is flexibility. Since the model is external, report creators cannot freely change every modelling element inside their report file. Live Connection is powerful when consistency and governance matter more than local model control.

Import	Direct Query	Live Connection
• Report performance is important • Scheduled refresh is enough • The dataset fits within capacity limits • The analyst needs modelling flexibility • The business does not need second-by-second updates • Power BI can safely store a copy of the data	• The data changes frequently • The dataset is too large to import • The organisation does not want to move the data • Security rules must remain enforced at the source • Near real-time visibility is more important than full modelling flexibility	• A trusted semantic model already exists • Business logic needs to remain centralised • Multiple reports should use the same definitions • Measures, relationships, and security rules are already managed upstream • Report creators should build visuals without duplicating the model

A practical way to decide:



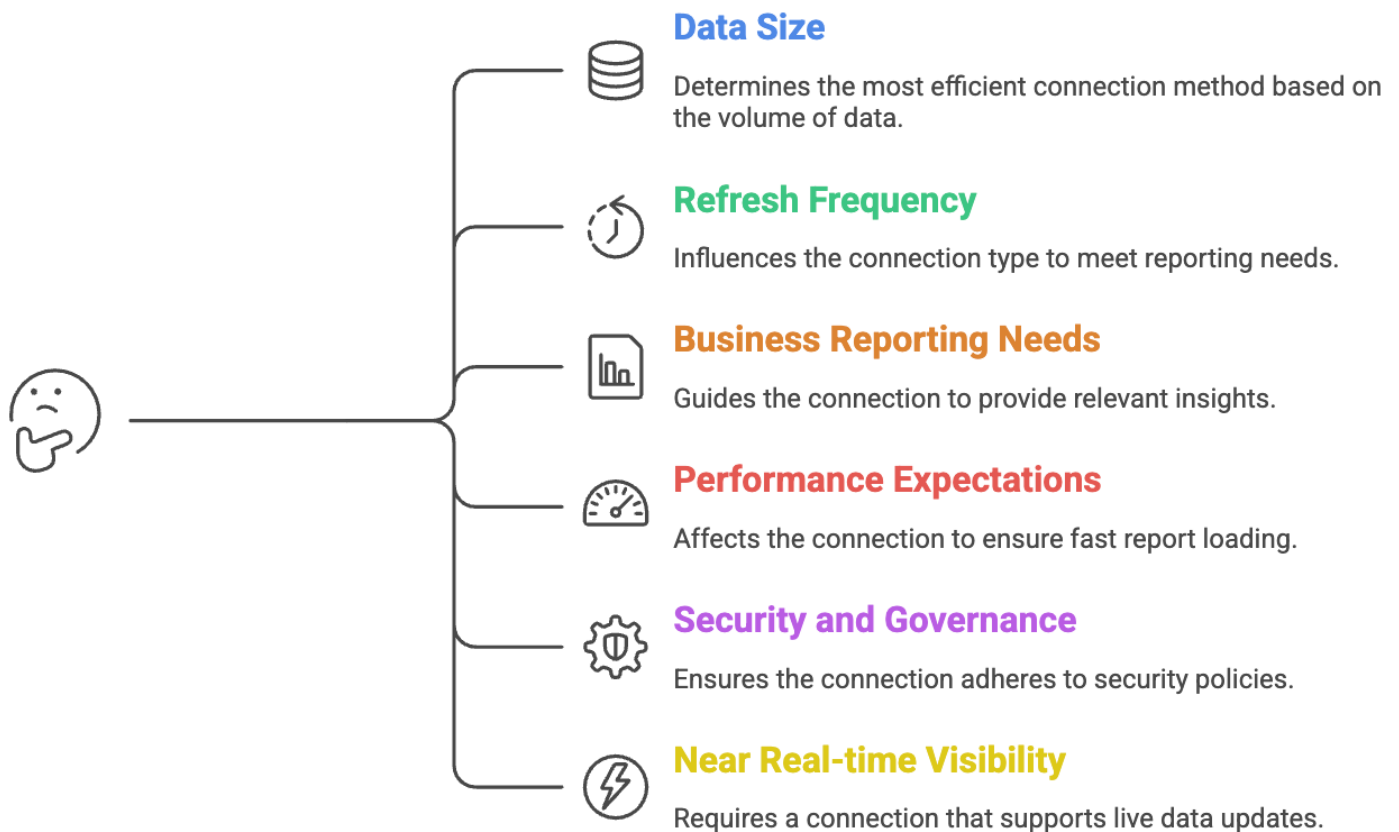
When to Use Each Data Connectivity Mode



Power BI

What Analysts Should Ask Before Connecting Data

How should Power BI connect to the data?



Conclusion

Data connection is more than the first technical step in Power BI. It shapes the performance, reliability, flexibility, and governance of the entire reporting solution.

A strong analyst does not only ask, "Where is the data?" They also ask, "What is the best way for Power BI to connect to this data for the business need?"

Choosing the right connection method early helps prevent slow reports, refresh issues, duplicated models, and inconsistent business logic later.

Stay tuned for next week's article in [The Insight Stack](#) : **Power BI, Layer by Layer**, where we move into **Week 3: Power Query and Data Preparation** and explore how analysts clean, shape, and prepare raw data before modelling and reporting.

Further Reading:

Further Reading:

- Types of Data Connection: [DirectQuery in Power BI: When to Use, Limitations, Alternatives - Power BI | Microsoft Learn](#)
- Difference between Live Connection and DirectQuery: [What's the difference between live connections and DirectQuery? - Power BI | Microsoft Learn](#)
- Power BI Data Sources: [Power BI data sources - Power BI | Microsoft Learn](#)
- Power Query Connectors: [List of all Power Query connectors - Power Query | Microsoft Learn](#)
- Connecting to Data in Power BI Desktop: [Quickstart: Connect to data in Power BI Desktop - Power BI | Microsoft Learn](#)
- Data Refresh : [Data refresh in Power BI - Power BI | Microsoft Learn](#)